

**Unit ID: 675d7732ea0163e7e5195af6**

## **Unit 3 - Repetition Statements**

### **PDF Documents**

**PDF Document 1: 1734178610280-Chapter 3 - unit 3 - Repetition Statements.pdf**

This PDF document is included in the unit materials.

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## UNIT 3 REPETITION STATEMENTS

### WHILE STATEMENT

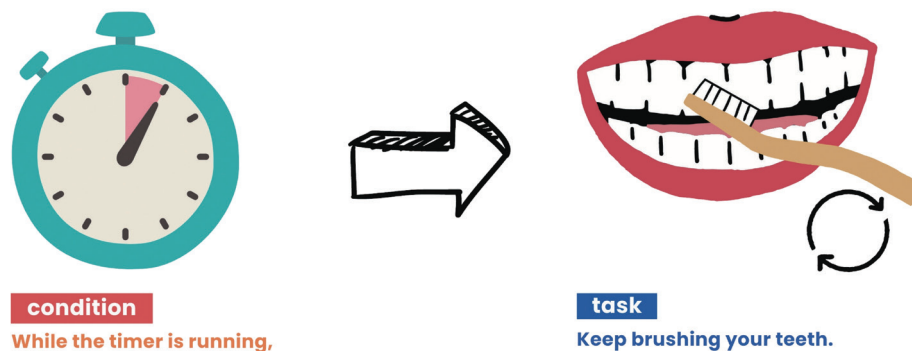
A "while" loop continues to execute its block of code **as long as the condition is true**. It checks the condition before running the loop's code. If the condition is true, the loop runs. If it's false, the loop stops. For example, "while the sun is up, keep the LED off."

### ✓ WHILE



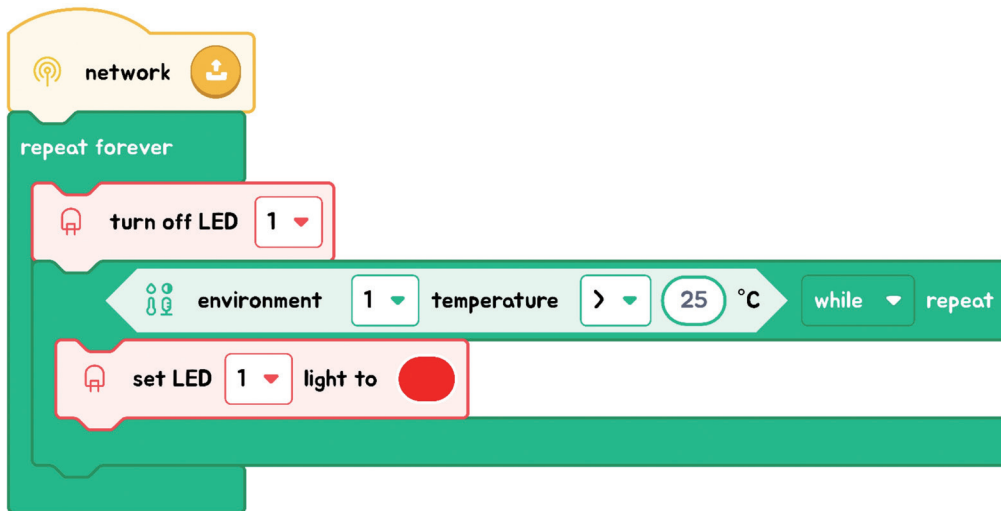
### Another Example:

"Think of using a 'while' loop like brushing your teeth while a timer is running. You start brushing, and 'while' the timer hasn't beeped (meaning time is not up yet), you keep brushing. As soon as the timer beeps, signaling the time is up, you stop brushing."





## EXPLANATION WITH CODE EXAMPLE



In our code example, we're working with a light sensor and an LED. When we start, our LED light is turned off. But we have a special rule written in our code about when to turn it on and make it yellow.

### 1) Is the illuminance level below 30%?

This checks how bright the light around us is. If it's really dim, less than 30% bright, our rule is met.

### 2) Set the LED to yellow.

Once we know the light is dim enough, our code tells the LED to turn on and shine in a yellow color. This makes the LED light up yellow as long as the room stays dim.

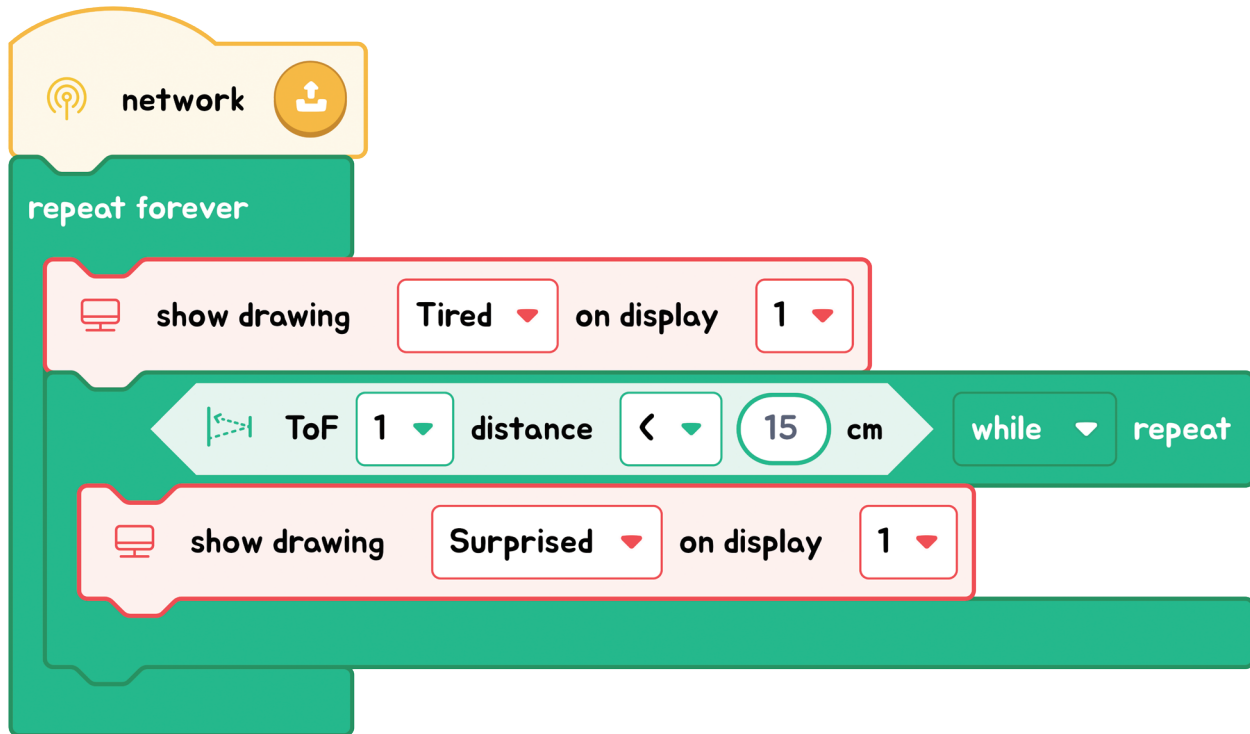


#### Code Insight

1. **Default State:** Initially, the LED is off. This state persists **until the "while" loop starts and when the luminance condition isn't met.**
2. **Condition Checking:** The code first checks **if the light level is below 30%,** determining the next steps.
3. **Execution While True:** With light levels under 30%, the LED turns yellow, repeating this task as long as the dim condition lasts.
4. **Exiting the Loop:** **When light levels exceed 30%, the loop ends,** and the LED shuts off, awaiting the next dim condition.

### Example

- 1 We are spooking a tired MODI, when we are near by the MODI, it's surprised. Find the condition and the action(task) in the code.



- 2 What is the purpose of a while loop in programming?
- (A) To execute a block of code only if a certain condition is true
  - (B) To repeatedly execute a block of code as long as a certain condition is true
  - (C) To execute a block of code regardless of any condition
  - (D) To execute a block of code once and then exit the loop

CORRECT ANSWER

(B)

- 3 When does a while loop stop executing?

- (A) When the condition inside the loop is false
- (B) When the condition inside the loop is true
- (C) When the loop encounters an error
- (D) When the loop reaches a specific line number

CORRECT ANSWER

(A)